



Geon Park

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👤 Profile

Master's student with a strong foundation in mathematics and theoretical computer science, applying it to practical systems research in software fuzzing and static analysis. My work focuses on translating abstract models into efficient, reliable software, blending deep analytical thinking from my mathematics background with hands-on systems programming in C and OCaml. Passionate about solving complex quantitative problems through rigorous, algorithmic approaches.

🎓 Education

KAIST, Master's in Information Security

March 2024 — February 2026

- Research in **software engineering** and **programming languages**, focusing on fuzzing and static analysis.
- Enhanced the baseline fuzzing tool's performance by 1.8-3.6x in bug finding time through targeted static analysis and novel strategies to focus on predefined target points.
- Research paper currently submitted to a peer-reviewed conference.
- Supported by [National Research Foundation of Korea scholarship](#) of ₩12M (~\$8,300), Jul 2024 – Jun 2025.

KAIST, B.S. Mathematics & Computer Science (Double Major)

March 2017 — February 2024

- Explored applied statistics, numerical analysis, discrete mathematics, and graph theory.
- Studied graph theory in depth, focusing on theoretical foundations and computational applications.

🔑 Key competencies

- **Logical Thinking:** Built strong analytical foundation through early training for the Korean Mathematical Olympiad (KMO, Middle School level, Gold Award in the first round).
- **Programming:**
 - **Python:** Developed a team project on data crawling & analysis (detailed in Extracurricular activity section); Undergrad TA for CS101, guided beginners.
 - **C:** Proficient in C, applied in master's research on fuzzing.
 - **LLVM IR experience:** Used as benchmark programs for research.
 - **OCaml:** Developed static analyzers as a front-end for research, created assignments as a TA for static analyzer-building coursework, built auto-grading system with Dune.
 - **Rust:** Coursework experience, focused on safe concurrency and memory management.
- **Machine Learning:** Academic project on ML-based image restoration ([link](#)).
- **Communication:** Native Korean, fluent English (1.5 years working with U.S. Army + English-based TA roles).

🏆 Extra-curricular activities

Military Service at Korean Augmentation to the United States Army (KATUSA), Pyeongtaek, Korea

November 2020 — May 2022

- Worked in Information Management Office, managing IT assets and services.

- Awarded Army Commendation Medal (ARCOM) for dedication.
- Strengthened collaboration in English-based international environment.

Team Leader, Data Crawling & Analysis Project (Tech for Impact)

September 2023 — December 2023

- Participated in an industry-academia collaboration course organized by Tech for Impact (a Kakao-affiliated public-interest foundation)
- Served as team leader, managing a project to crawl data from local council websites
- Extracted and analyzed council member information, focusing on diversity metrics such as age and gender
- Processed data into a user-friendly format to enhance accessibility and transparency
- Project outcomes:
 - [Demo video \(Korean\)](#)
 - [Code repository \(Python-based, GitHub\)](#)
 - [Interview article featured in *Digital Times* \(Korean\)](#), where I discussed the initiative's impact

🦋 Skills

- Programming Languages & Tools: OCaml, Python, C, Rust
- Quantitative & Theoretical Foundation: Applied Statistics, Numerical Analysis, Graph Theory
- Languages: Korean (native), English (fluent)